

KATRIN BURKART, *M.Sc., Ph.D.*

1. PERSONAL DATA

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Residence: Seattle, WA, USA

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2. EDUCATION

Doctorate, <i>Dr. rer. nat.</i> (equiv. to Ph.D.)	Humboldt-Universität zu Berlin, Germany (<i>summa cum laude</i>) Concentrations: Climate and Health	Oct 2011
Diploma (equiv. to M.Sc.)	Humboldt-Universität zu Berlin, Germany (<i>with distinction</i>) Major: Geography, Minors: Meteorology, Chemistry	Apr 2007
Intermediate diploma (equiv. to B.Sc.)	Friedrich-Alexander Universität, Erlangen-Nürnberg, Germany Major: Geography, Minors: Chemistry, Geology	Sep 2003

3. POSTGRADUATE TRAINING

Research Scientist (with Darby Jack, Sc. D.) | **Columbia University** | **Mailman School of Public Health, Department of Environmental Health Sciences** | **New York, NY, USA**

Feb 2017 – Mar 2018

- In charge of data analyses and epidemiological modelling within Ghana randomized air pollution and health study (GRAPHS)
- Collaborated and coordinated research with several researchers within GRAPHS
- Contributed to research design of the New York City Bike Share Project
- Communicated research to national and international scientific communities
- Mentored graduate students

Postdoctoral Research Fellow (with Patrick L. Kinney, Sc. D.) | **Columbia University** | **Mailman School of Public Health, Department of Environmental Health Sciences** | **New York, NY, USA**

Feb 2015 – Jan 2017

- Conducted statistical modelling of heat, cold and precipitation health effects in Bangladesh
- Carried-out projections of temperature-related mortality in Bangladesh under climate change
- Initiated research on the potential of internet search engine and social media data for environmental health research
- Submitted three grant applications (two as PI, one as Co-PI)
- Established collaborations with researchers from the US (Earth Institute, International Institute for Climate and the Society, Columbia University) and Europe (French Institute of Public Health Surveillance, University of Umeå)

Postdoctoral Research Fellow (with Wilfried Endlicher, *Prof. Dr.*) | **Humboldt-Universität zu Berlin** | **Geography Institute, Department of Climatology** | **Berlin, Germany**

May 2011 – Dec 2014

- Research on mitigation of temperature effects through urban vegetation and water bodies
- Taught six undergraduate and two graduate courses
- Mentored four undergraduate and two graduate students
- Submitted two grant applications (one as PI, one as Co-PI)
- Coordinated research with partners from Germany (Universities of Bonn and Bielefeld, Helmholtz Gemeinschaft), the University of Lisbon and the Czech Academy of Sciences
- Contributed to academic services (member of doctoral committees and three search and screen committees, deputy member of Institute Council)

4. FACULTY POSITION HELD

Assistant Professor, Department of Health Metrics Science, University of Washington Since 2018

Adjunct Assistant Professor, Department of Environmental and Occupational Health Science, University of Washington Since 2021

Adjunct Assistant Professor, Department of Global Health Science, University of Washington Since 2018

5. HOSPITAL POSITION HELD

N/A

6. HONORS

Summa cum laude graduate distinction, Humboldt University of Berlin, 2011, Berlin, Germany

Travel award and scholarship to the student training course in ‘Capacity Building in Global Change and Health Research’. Beijing Workshop November 2009, Beijing, China

Travel award and scholarship to the International NCCR Climate Summer School: Key challenges in climate variability and change, August 31st to September 5th, 2008, Monte Verità, Switzerland

Advancement award for diploma thesis ‘Size-distribution of polycyclic aromatic hydrocarbons on atmospheric particles’ awarded by the ‘Institut für Agrar- und Stadtökologische Projekte’ [Institute of agriculture and urban ecology], Berlin, Germany

7. BOARD CERTIFICATION:

N/A

8. CURRENT LICENSE(S) TO PRACTICE

N/A

9. DIVERSITY, EQUITY AND INCLUSION ACTIVITIES

- Diversity and Inclusion Committee at the Department of Health Metrics Science
- Mentorship of a diverse student body and young professionals
- Publications investigating and highlighting public health issues from an equity perspective

- Regular participation in DEI discussion at the Institute for Health Metrics
- Integration of DEI topics into teaching and mentorship

10. PROFESSIONAL ORGANIZATIONS:

International Society for Environmental Epidemiology (*ISEE*)

International Society for Pharmacoepidemiology (*ISPE*)

11. EDUCATION AND TRAINING ACTIVITIES

A. DIDACTIC TEACHING

Humboldt-Universität zu Berlin, Geography Institute, Berlin, Germany

- **Urban health and the built environment** (German title: Stadtgesundheit und urbaner Lebensraum), undergraduate seminar, primary instructor in cooperation with Tobia Lakes, block seminar taught in summer 2015
- **Bioclimate and health – time-series analysis and statistical modeling for climate and health data** (German title: Bioklima und Gesundheit – Zeitliche Analyse und Modellierung von Klima und Gesundheitsdaten), undergraduate class, primary instructor, weekly course taught in summer 2014
- **A web-based GIS for health and the environment – statistical and GIS-based approaches** (German title: Ein WebGIS zur Umwelt und Gesundheit in Berlin - Statistische und GIS-basierte Methoden), graduate course, primary instructor in cooperation with Tobia Lakes, weekly course taught in summer 2014
- **Advanced terrain climatology** (German title: Geländeklimatologie für Fortgeschrittene), graduate course, primary instructor, weekly course taught and 1 week field work in summer 2014
- **Introduction to Climatology** (German title: Einführung in die Klimatologie), undergraduate class, primary instructor, weekly class taught in summer 2013 and 2014
- **Analyzing geodata in health and environmental research using R and QGIS** (German title: Geodatenanalyse in der Mensch-Umwelt-Systemforschung mit R und QGIS), graduate seminar, primary instructor, 2-day block seminar in cooperation with Tobia Lakes taught in fall 2015
- **Scientific programming with R (German title: Wissenschaftliche Programmierung)**, undergraduate course, primary instructor, weekly class taught in winter 2013/14

University of Bielefeld, School of Public Health, Germany

- **Climate change and spatial epidemiology. Using R for basic health statistics and data modelling.** Taught at the 5th International Summer School on ‘Spatial Epidemiology, Climate and Health: Concepts and Modelling’, University of Bielefeld: September 2013

University of Washington, Department of Health Metrics/ Department of Global Health, Seattle, WA

- G H 592 A: Health Metrics Science seminar (co-taught with Dr. David Pigott winter spring and autumn 2019 and winter 2020)

University of Washington, School of Medicine, Department of Health Metric Science

- HMS 530: Global Burden Of Disease: Methods And Results (Spring 2024-2026, co-taught with Dr. Maegan Dirac)

University of Washington, School of Public Health, Health Systems and Population Health

- GH 511: Problems in Global Health (Summer 2022-2024, Winter 2023-2025)

University of Washington, Information School

- INFO 478 Population Health Informatics (Spring 2021, Winter 2022)

University of Washington, Institute of Health Metrics Science

- GBD Learning Week, November 2025 „GBD 2023 results and impacts“
- GBD Foundations Workshop 2026

B. THESIS COMMITTEE MEMBERSHIP

- Sarah Osenberg: **Personal heat exposure measurements using temperature sensors, GPS and an activity diary** (German title: Individuelle Messungen von Hitzestress mittels Temperaturmessungen, GPS und Aktivitätstagebuch – ein Methodendesign). Master thesis co-supervised with Tobia Lakes. Thesis submitted at the Humboldt-Universität zu Berlin (*M.Sc. thesis*)
- Anne Plesse: **Relationship between body weight and socio-cultural determinants – Spatial analysis of school enrollment data in Berlin** (German title: Zusammenhänge zwischen Körpergewicht und soziokulturellen Faktoren – Räumliche Betrachtung der Berliner Einschulungsdaten nach Prognoseräumen von 2006 – 2012). Bachelor thesis co-supervised with Tobia Lakes. Thesis submitted at the Humboldt-Universität zu Berlin (*M.Sc. thesis*)
- Gregor Qualitz: **Effects of El-Niño Southern Oscillation Events on the Tuna Fisheries in the Western Central Pacific Oceans**. Co-supervised with Wilfried Endlicher. Bachelor thesis submitted at the Humboldt-Universität zu Berlin (*M.Sc. thesis*)
- Isabella Stingl and Susan Karlebowsky: **Interim use of inner-city brown fields – Challenges and potentials for implementation strategies** (German title: Zwischennutzung innerstädtischer Brachflächen – Potentiale, Hemmnisse und Umsetzungsstrategien auf Grundlage ökologischer Daten und einer Bürgerbefragung ausgeführt am Beispiel der Cuvrybrache). Bachelor thesis co-supervised with Wilfried Endlicher. Thesis submitted at the Humboldt-Universität zu Berlin (*M.Sc. thesis*)
- Sarah Wozniak. **Estimating the impact of temperature on cause-specific mortality using Distributed Lag non-linear Model**. Thesis submitted at the University of Washington (*MPH thesis*)
- William Godwin: **Assessment of interactive effects of temperature and air pollution on mortality in Mexico City**. Thesis submitted at the University of Washington (*MPH thesis*)
- Susan Campbell. **Effect of projected flood exposures under different climate change scenarios on population-level mortality outcomes**. Capstone project towards completion of M.S. in Health Metrics Sciences, University of Washington (*Capstone*)

- Chengyi Lin. **Hypothetical Ambient Fine Particulate Matter Interventions and Dementia Risk: Causal Effect and the Role of Cardiovascular Disease** (*PhD committee*)
- Flávia Araújo Girardi. **The burden of Lung Cancer and mortality attributable to occupational risk factors in Brazil and other countries: estimates from the Global Burden of Disease Study 2021.** Doctoral research project conducted at the Institute for Health Metrics Science (*PhD committee*)

C. ADVISING AND MENTORSHIP

- **Sarah Wozniak** – Research Supervisor; focused on air pollution modeling and Global Burden of Disease (GBD) estimations.
- **Will Godwin** – Research Supervisor; focused on air pollution modeling and Global Burden of Disease (GBD) estimations.
- **Nikki DeCleene** – Faculty Advisor for Master of Science candidacy.
- **Xinmei Huang** – PhD Rotation Supervisor; led research on the impact of air pollution on dementia and burden-of-proof analysis for stroke and ischemic heart disease (IHD).
- **Flávia Araújo Girardi, PhD** – Supervisor for the systematic assessment of the burden of proof regarding the carcinogenic effects of asbestos.
- **Mehnaz Munir, PhD** – Supervisor for burden-of-proof assessments for $\text{PM}_{2.5}$ impacts on COPD, lower respiratory infections, and lung cancer.
- **Yasna Palmeira Silva, PhD** – Advisor for research estimating the impacts of heat in South America, with a specific focus on Chile.

12. . EDITORIAL RESPONSIBILITIES

None to date

13. SPECIAL RESPONSIBILITIES AND SERVICES

a) International

- Official partner of the **WHO-WMO Global Heat Health Information Network (GHHIN)** representing IHME (invited by Dr. Yoonhee Kim and Alejandro Saez Reale)
- Expert with the global network '**Global Heat Health Information Network**', supported by **WHO and WMO** (invited by Alejandro Saez Reale and Maddie West)
- **Forethought Foundation**, Consultation on Good Judgment's Superforecasters (invited by Drs. Luis Enrique Urtubey and John Halstead)
- **IPCC, 6th Assessment Report**: Health, wellbeing and the changing structure of communities (contributing author invited by Dr. Jeremy Hess)
- Consultant **CS Publishing**, supporting book development on climate adaptation (Title: „Eis gegen Heiß“)

- Consulting the **Robert Koch Institute** on national Burden of Climate Change estimations for Germany (invited by Dr. Aline Anton)

b) National

- **NIH/CRS: NIH Social Sciences and Population Studies Study Section A [SSPA]** review group (invited by Dr. Suzanne Ryan)
- **NIH/NIEHS: R21 Application for Time-Sensitive Research Opportunities in EHS – Special Emphasis Panel (SEP)** Review Meeting, September 2022 (invited by Dr. Quentin Li)
- **REACH - Research and Engagement on Adaptation for Climate and Health** (invited by Prof. Dr. Jeremy Hess)

c) Regional/University of Washington

- **Policy advisory council**, supporting the effective translation, communication, and dissemination of research findings on NIH-funded project “Effects of Extreme Temperature on Aging Populations in the United States”
- **Diversity and Inclusion Committee** at the Department of Health Metrics Science, University of Washington, founding member, August 2021 – ongoing
- **PhD selection committee** at Department of Health Metrics, University of Washington in winter 2019 and 2021, 2022, and 2024
- **M.Sc. selection committee** at Department of Health Metrics, University of Washington in winter 2021
- **MPH selection committee** at Department of Health Metrics, University of Washington in winter 2019 and 2020
- **Diversity and inclusion committee** at Department of Health Metrics, University of Washington in winter 2019

14. CLINICAL ACTIVITIES:

N/A

RESEARCH FUNDING

A. ACTIVE

Pfizer, Initiative Fund

03/01/2026-07/31/2027

Estimating the Global Incidence of RSV-Related Hospitalizations in High-Risk Populations: Global Modeling

This grant initiative supports estimations RSV-attributable respiratory and cardiorespiratory hospitalizations for 204 countries and territories. The project builds upon prior work conducted by the PI, Dr. Katrin Burkart and uses country-level estimates of RSV-attributable hospitalizations in high-risk groups to build a global model driven by covariates.

Role: Principal Investigator

Budget: \$920 K

National Institute of Aging (R01AG084536)

01/01/2025 – 12/31/2029

Effects of Extreme Temperature on Aging Populations in the United States: A Geographically Granular Assessment of Present and Future Impacts by Race and Ethnicity

The research project aims to estimate the current and future burden of extreme temperatures by county across the entire U.S. by integrating detailed geographic, socio-demographic, and economic data. It will assess risks on a granular scale, project future impacts under climate and population change scenarios, and evaluate intervention strategies to mitigate temperature-related health risks. The study leverages extensive IHME data and infrastructure, with results to be disseminated through scientific publications and user-friendly visualization tools for researchers, policymakers, and the public.

Role: Principal Investigator**Budget: \$3.8 million**

Pfizer, Initiative Fund

01/01/2025-04/31/2026

Estimating the incidence of RSV-attributable (cardio)respiratory hospitalizations among high-risk patients in high- and middle-income countries

This grant initiative supports estimations of RSV-attributable respiratory and cardiorespiratory hospitalizations. The project leverages the Comparative Risk Assessment framework (CRA) and builds upon prior work. It combines individual level clinical data with population level exposure data and derives exposure-response relationships between RSV activity and cardio-respiratory hospitalizations.

Role: Principal Investigator**Budget: \$800 K**

Bill & Melinda Gates Foundation (OPP1152504)

Murray (PI)

07/01/2017–06/30/2027

Global Public Goods: Global Burden of Disease (GBD), Forecasting, and Resource Tracking

This grant supports IHME's ongoing work to expand and consistently update the high quality, scientific, quantitative evidence base for policymakers, donors, researchers, and other decision-makers in order to improve population health. The focus of this grant is quantifying the burden of disease, financing for global health, and forecasting and future scenarios of health outcomes.

Role: Key Personnel

Pfizer, Initiative Fund

01/01/2025-04/31/2026

Estimating the global incidence of RSV-related hospitalizations

This grant initiative supports estimations of global RSV-attributable respiratory and cardiorespiratory hospitalizations for 204 countries from 2015 to 2019. The project builds upon a previous project and generates a universal exposure-response relationship using a metaregression framework developed at IHME. Global time-series of weekly RSV-exposures were generated using RSV surveillance data and a spatio-temporal, i.e. RegMod. Global exposures will be combined with exposure-response relationships using the Comparative Risk Assessment framework, an integral part of the Global Burden of Disease framework.

Role: Principal Investigator**Budget: \$900 K**

Pfizer, Initiative Fund

01/01/2025-04/31/2026

Estimations of hospitalizations and Incidence rates of Clostridioides difficile (C. diff) in selected locations*This funding roposal outlines estimates of incidence rates and hospitalizations initiative supports***Role: Principal Investigator****Budget: \$760 K****B. COMPLETED**

Pfizer, Initiative Fund

01/01/2025-04/31/2026

RSV-attributable hospitalizations among adults in select high- and middle-income countries*This grant initiative supports estimations of RSV-attributable respiratory and cardiorespiratory hospitalizations in select high- and middle-income countries. The project leverages the Comparative Risk Assessment framework (CRA) and combines individual level clinical data with population level exposure data from RSV surveillance system. Derived exposure-response relationships between RSV activity and cardio-respiratory hospitalizations are combine with exposure information in order to calculate the Population Attributable Fraction and the attributable hospitalizations.***Role: Principal Investigator****Budget: \$1 million**

Health Effects Institute (4977/20-11-2)

06/01/2020-01/31/2022

Integrating satellites, ground monitoring, and modeling to estimate long-term NO2 exposures and associated pediatric asthma impacts (HEI-GWU_NO2 EXPOSURE)*This grant supports work on evaluating the evidence of the impact of NO2 and pediatric asthma including evidence scoring using the IHME internal scoring system. Furthermore, the grant supports the integration of NO2 and asthma into the Global Burden of Disease Study***Role: Principal Investigator**

Initiative Fund, Institute for Health Metric Science

01/01/2021-03/31/2022

Client Service Unit - Estimating contributions of influenza to ischemic heart disease mortality*This grant initiative supports estimations of the influenza attributable burden of ischemic heart disease. Key components of the work include the derivation of exposure response relationships between influenza activity and ischemic heart disease as well as global estimations of the attributable burden for 204 countries and territories.***Role: Investigator**

National Institute of Aging (P30 AG047845)

Murray (PD)

04/01/2017 – 06/30/2018

UW Center for Demography and Economics of Aging*The Demography Center seeks to facilitate innovative social science investigations on the determinants of population health that leverage the substantial investments in the maintenance and expansion of the GBD database to improve healthy aging by populations around the world. This pilot project's main aim is to estimate costs, decompose changes, and forecast future spending on Alzheimer's disease and other dementias in the US***Role: Key Personnel**

German Research Foundation (BU 3134/1-2)

Burkart (PI)

08/01/2016-01/31/2017

Projecting temperature-related mortality in Bangladesh under climate change scenarios – considering cold effects and precipitation

Aim of this project was to improve projections of climate-change related impacts on mortality in Bangladesh. Special consideration was given to estimating cold-related mortality under global warming. Outcomes indicated that reduced cold effects do not offset increases in heat-related mortality leading to an overall increase in temperature-related mortality due to climate change. The relationship between precipitation and mortality was highly complex varying by season, age and gender complicating realistic estimates of future impacts under climate change and highlighting a strong research need.

Role: Principal Investigator

German Research Foundation (BU 3134/1-1)

Burkart (PI)

02/01/2015 - 07/31/2016

Projecting temperature-related mortality in Bangladesh under climate change scenarios - with consideration of ageing, urbanization, air pollution and epidemiological transition.

The goal of this project was to provide future projections of temperature-related mortality in Bangladesh. Dose-response relationships between temperature and mortality were estimated for different locations and population groups and linked to future projections of climate stemming from the NEX-GDDP multi-model data set. Findings indicated substantial increases in heat-related mortality, especially in urban areas and for the elderly.

Role: Principal Investigator**15. BIBLIOGRAPHY****A. JOURNAL ARTICLES (PEER-REVIEWED)****2009**

1. **Burkart K**, Endlicher W. Assessing the atmospheric impact on public health in the megacity of Dhaka, Bangladesh. *Erde* 2009; **140**.
2. **Burkart KG**, Gruebner, O, MMH Khan, Staffeld R. Megacity Dhaka: Informal settlements, urban Environment and public Health. *Geographische Rundschau* 2009; **1**.

2011

3. **Burkart K**, Khan MH, Krämer A, Breitner S, Schneider A, Endlicher WR. Seasonal variations of all-cause and cause-specific mortality by age, gender, and socioeconomic condition in urban and rural areas of Bangladesh. *International Journal for Equity in Health* 2011; **10**. PMID: 21816075
4. **Burkart K**, Schneider A, Breitner S, Khan MH, Krämer A, Endlicher W. The effect of atmospheric thermal conditions and urban thermal pollution on all-cause and cardiovascular mortality in Bangladesh. *Environmental Pollution* 2011; **159**. PMID: 21377776
5. Gruebner O, Staffeld R, Khan M, **Burkart K**, Krämer A, Hostert P. Urban health in megacities: Extending the framework for developing countries. *Magazine of the International Human Dimensions Programme on Global Environmental Change* 2011; **1**: 40–7.

2013

6. Cheong YL, **Burkart K**, Leitão PJ, Lakes T. Assessing weather effects on dengue disease in Malaysia. *Int J Environ Res Public Health* 2013; **10**: 6319–34. PMID: 24287855
7. **Burkart K**, Nehls I, Win T, Endlicher W. The carcinogenic risk and variability of particulate-bound polycyclic aromatic hydrocarbons with consideration of meteorological conditions. *Air Quality, Atmosphere and Health* 2013; **6**.
8. **Burkart K**, Canário P, Breitner S, *et al.* Interactive short-term effects of equivalent temperature and air pollution on human mortality in Berlin and Lisbon. *Environmental pollution* 2013; **183**. PMID: 23941745

2014

9. Schuster C, **Burkart K**, Lakes T. Heat mortality in Berlin – Spatial variability at the neighborhood scale. *Urban Climate* 2014; **10**: 134–47.
10. Khan M, Zanzudana A, **Burkart K**, Krämer A. Determinants of Diarrhoea in ‘Urban’ Slums of Dhaka and Adjacent Rural Areas: a Household-level Analysis. *Population, Space and Place* 2014; **20**: 498–511.
11. **Burkart K**, Khan MMH, Schneider A, *et al.* The effects of season and meteorology on human mortality in tropical climates: A systematic review. *Transactions of the Royal Society of Tropical Medicine and Hygiene* 2014; **108**. PMID: 24907712
12. **Burkart K**, Breitner S, Schneider A, Khan MMH, Krämer A, Endlicher W. An analysis of heat effects in different subpopulations of Bangladesh. *International Journal of Biometeorology* 2014; **58**. PMID: 23689928

2015

13. Thiele-Eich I, **Burkart K**, Simmer C. Trends in Water Level and Flooding in Dhaka, Bangladesh and Their Impact on Mortality. *Int J Environ Res Public Health* 2015; **12**: 1196–215. PMID: 25648177

2016

14. Lakes T, **Burkart K**. Childhood overweight in Berlin: intra-urban differences and underlying influencing factors. *International Journal of Health Geographics* 2016; **15**: 12. PMID: 27001085
15. Urban A, **Burkart K**, Kysely J, *et al.* Spatial Patterns of Heat-Related Cardiovascular Mortality in the Czech Republic. *Int J Environ Res Public Health* 2016; **13**: 284. PMID: 26959044
16. **Burkart K**, Kinney P. Is precipitation a predictor of mortality in Bangladesh? A multi-stratified analysis in a South Asian monsoon climate. *Science of the Total Environment* 2016; **553**. PMID: 26933968
17. **Burkart K**, Meier F, Schneider A, *et al.* Modification of heat-related mortality in an elderly urban population by vegetation (Urban green) and proximity to water (Urban blue): Evidence from Lisbon, Portugal. *Environmental Health Perspectives* 2016; **124**. PMID: 26566198

2017

18. Gruebner O, Khan M, **Burkart K**, *et al.* Spatial variations and determinants of infant and under-five mortality in

Bangladesh. *Health Place* 2017; **47**: 156–64. PMID: 28888890

19. Nissan H, **Burkart K**, Perez EC de, Aalst MV, Mason S. Defining and Predicting Heat Waves in Bangladesh. 2017; published online Oct 1. DOI:10.1175/JAMC-D-17-0035.1.
20. **Burkart K**, Kinney PL. What drives cold-related excess mortality in a south Asian tropical monsoon climate—season vs. temperatures and diurnal temperature changes. *International Journal of Biometeorology* 2017; **61**. PMID: 27995322

2019

21. Asante K, Wylie BJ, , ... **Burkart KG**, *et al.* The Ghana Randomized Air Pollution and Health Study (GRAPHS): a cluster randomized trial of Liquified petroleum gas (LPG) and efficient biomass cookstoves delivered during pregnancy. *Environmental Epidemiology* 2019; **3**: 296. PMID: 26395578
22. Kaali S, Jack DW, Delimini R, ... **Burkart KG**, *et al.* Prenatal Household Air Pollution Alters Cord Blood Mononuclear Cell Mitochondrial DNA Copy Number: Sex-Specific Associations. *Int J Environ Res Public Health* 2019; **16**: 26. PMID: 30583542
23. Heaney AK, Carrión D, **Burkart K**, Lesk C, Jack D. Climate Change and Physical Activity: Estimated Impacts of Ambient Temperatures on Bikeshare Usage in New York City. *Environ Health Perspect* 2019; **127**: 037002. PMID: 30835141
24. Lee AG, Kaali S, Quinn A, ... **Burkart KG**, *et al.* Prenatal Household Air Pollution Is Associated with Impaired Infant Lung Function with Sex-Specific Effects. Evidence from GRAPHS, a Cluster Randomized Cookstove Intervention Trial. *Am J Respir Crit Care Med* 2019; **199**: 738–46. PMID: 30256656
25. Sagnik Dey, Gupta T, R S Dhaliwal, **Burkart KG**, Dandonda L. The impact of air pollution on deaths, disease burden, and life expectancy across the states of India: the Global Burden of Disease Study 2017. *Lancet Planet Health* 2019; **3**: e26–39. PMID: 30528905

2020

26. Murray CJL, Aravkin AY, Zheng P, ... **Burkart KG**, *et al.* Global burden of 87 risk factors in 204 countries and territories, 1990–2019: a systematic analysis for the Global Burden of Disease Study 2019. *The Lancet* 2020; **396**: 1223–49. PMID: 33069327
27. Vos T, Lim SS, Abbafati C, ... **Burkart KG**, *et al.* Global burden of 369 diseases and injuries in 204 countries and territories, 1990–2019: a systematic analysis for the Global Burden of Disease Study 2019. *The Lancet* 2020; **396**: 1204–22. PMID: 33069326

2021

28. Nichols Emma, Foad Abd-Allah, Amir Abdoli, ... **Burkart KG**, *et al.* Global mortality from dementia: Application of a new method and results from the Global Burden of Disease Study 2019. *Alzheimers Dement (N Y)* 2021; **7**: e12200. PMID: 34337138
29. Ghosh R, Causey K, **Burkart KG**, Wozniak S, Cohen A, Brauer M. Ambient and household PM2.5 pollution and adverse perinatal outcomes: A meta-regression and analysis of attributable global burden for 204 countries and

territories. *PLoS Med* 2021; **18**: e1003718. PMID: 34582444

30. Nichols E, Abd-Allah F, Abdoli A, ... **Burkart KG**, et al. Use of multidimensional item response theory methods for dementia prevalence prediction: an example using the Health and Retirement Survey and the Aging, Demographics, and Memory Study. *BMC Medical Informatics and Decision Making* 2021; **21**: 241. PMID: 34380485
31. **Burkart KG**, Brauer M, Aravkin AY, ... **Burkart KG**, et al. Estimating the cause-specific relative risks of non-optimal temperature on daily mortality: a two-part modelling approach applied to the Global Burden of Disease Study. *The Lancet* 2021; **398**: 685–97. PMID: 34419204
32. Causey K, **Burkart KG**, Collman G, Hu H, Brauer M, London SJ. Increasing the Impact of Environmental Epidemiology in the Global Burden of Disease Project. *Epidemiology* 2021; **32**: 1–5. PMID: 33031218
33. Kendrick PJ, Reitsma MB, Abbasi-Kangevari M, ... **Burkart KG**, et al. Spatial, temporal, and demographic patterns in prevalence of chewing tobacco use in 204 countries and territories, 1990–2019: a systematic analysis from the Global Burden of Disease Study 2019. *The Lancet Public Health* 2021; **6**: e482–99. PMID: 34051920
34. Fablina Sharara*, Eve E Wool*, Gregory J Bertolacci, ... **Burkart KG**, et al. Fatal police violence by race and state in the USA, 1980–2019: a network meta-regression. *The Lancet* 2021; **398**: 1239–55. PMID: 34600625
35. Chillrud SN, Ae-Ngibise KA, Gould CF, ... **Burkart KG**, et al. The effect of clean cooking interventions on mother and child personal exposure to air pollution: results from the Ghana Randomized Air Pollution and Health Study (GRAPHHS). *J Expo Sci Environ Epidemiol* 2021; **31**: 683–98. PMID: [33654272](#)
36. Quinn AK, Adjei IA, Ae-Ngibise KA, ... **Burkart KG**, et al. Prenatal household air pollutant exposure is associated with reduced size and gestational age at birth among a cohort of Ghanaian infants. *Environ Int* 2021; **155**: 106659. PMID: 34134048
37. Paulson KR, Kamath AM, Alam T, ... **Burkart KG**, et al. Global, regional, and national progress towards Sustainable Development Goal 3.2 for neonatal and child health: all-cause and cause-specific mortality findings from the Global Burden of Disease Study 2019. *The Lancet* 2021; **398**: 870–905.
38. Reitsma MB, Flor LS, Mullany EC, Gupta V, Hay SI, ... **Burkart KG**, et al. Gakidou E. Spatial, temporal, and demographic patterns in prevalence of smoking tobacco use and initiation among young people in 204 countries and territories, 1990–2019. *The Lancet Public Health* 2021; **6**: e472–81. PMID: 34051921
39. Haile LM, Kamenov K, Briant PS, ... **Burkart KG**, et al. Hearing loss prevalence and years lived with disability, 1990–2019: findings from the Global Burden of Disease Study 2019. *The Lancet* 2021; **397**: 996–1009. PMID: 33714390
40. Pandey A, Brauer M, Cropper ML, ... **Burkart KG**, et al. Health and economic impact of air pollution in the states of India: the Global Burden of Disease Study 2019. *The Lancet Planetary Health* 2021; **5**: e25–38. PMID: 33357500
41. Jahagirdar D, Walters MK, Novotney A, ... **Burkart KG**, et al. Global, regional, and national sex-specific burden and control of the HIV epidemic, 1990–2019, for 204 countries and territories: the Global Burden of Diseases Study 2019. *The Lancet HIV* 2021; **8**: e633–51. PMID: 34592142

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84. **Burkart, KG.**, Brauer, M., Roth, G.: Environment and the Heart: Measuring the health impact of environmental risks and making it matter. *Accepted at the Journal of the American College of Cardiology (JACC)*.

B. SUBMITTED MANUSCRIPTS

85. **Burkart, KG.**,, Reiner, B.: Forecasts of global, regional, and national disease burden from floods and tropical storms through 2100: a systematic analysis across climate scenarios. *Submitted to The Lancet*.
86. **Burkart, K.**, ..., Brauer, M.: Forecasts of global, regional and national burden of disease of wildfire and non-wildfire ambient particulate matter air pollution: a systematic analysis of climate scenarios to 2100. *Submitted to The Lancet*.
87. Stanaway, J., **Burkart, K.**, et al. Forecasts of global, regional, and national disease burden of non-optimal temperature: a systematic analysis across climate scenarios to 2100. *Submitted to The Lancet*.
88. Girardi, F.,, **Burkart, K.** Burden of cancer attributable to occupational asbestos exposure in the Americas, 1990–2023: an analysis using the Global Burden of Disease Study 2023. *Submitted to the Lancet Regional Health – The Americas*.
89. **Burkart, K.**, Rafferty, Q., Liang, C. et al.: Global, regional, and national estimates of adult respiratory and cardiorespiratory hospitalizations attributable to respiratory syncytial virus from 2015 to 2022: application of the Global Burden of Disease framework. *Submitted to the Lancet Global Health*.

C. BOOK CHAPTERS

1. **Burkart, K.**, Endlicher, W. (2011): Human Bioclimate and Thermal Stress in the Megacity of Dhaka, Bangladesh – A Climatological Approach to Health Relevance Assessment. In: Krämer A, Khan MH, Kraas F (Eds.). Springer. Heidelberg
2. Bruns, A., Blackburn, S., Perez Bolde, C.C., **Burkart, K.**, Francis, R., Kraft, D., Li, Y., Lu, Q., Olaniyi, A., Zhou, X. (2013): Environmental impacts of megacities on the coast. In: Pelling, M. & S. Blackburn, Urbanisation and the Coast. IGBP Synthesis Report. Routledge, Earthscan
3. Endlicher, W., Scherber, K., Koppe, C., **Burkart, K.**, (2015): Gesundheitliche Schäden durch thermische Belastung [Health hazards from thermal pollution]. In: Brasseur, G., Becker, P., Claußen, M., Hansjürgens, B., Kottmeier, C., Latif, M., Lemke, P., Mahrenholz, P., Merz, B., Sauerborn, R., Schellnhuber, H.-J., Vereecken, H. (Hrsg.): Klimawandel und Anpassung: Multidisziplinärer Bericht zu Deutschlands Zukunft [Climate change and adaptation: Multidisciplinary Report on the Future of Germany]. Climate Service Center Germany, Hamburg, Germany

D. THESES

1. **Burkart, Katrin:** Größenverteilung polyzyklisch atmosphärischer Kohlenwasserstoffe an atmosphärischen Partikeln. Unveröffentlichte Diplomarbeit [Size distribution of polycyclic aromatic hydrocarbons associated to atmospheric particles]. Eingereicht an der Math.-Nat. Fakultät der Humboldt-Universität zu Berlin. Geographisches Institut. 12/2006
2. **Burkart, Katrin:** Bioclimate and health in urban and rural areas of Bangladesh – short- and long-term effects of atmospheric thermal conditions on human mortality; Dissertation, Humboldt-Universität zu Berlin, Mathematisch-Naturwissenschaftliche Fakultät II, publiziert am 07.10.2011

16. INVITED TALKS

A. NATIONAL

1. **The mitigation effect of urban green on heat-related excess mortality in Lisbon, Portugal.** Environmental Health Conference. March 2013, Boston, MA, USA.
2. **Effects of weather and climate on mortality in different subpopulation – the role of atmospheric and non-atmospheric modifiers.** Invited Seminar: Climate and Health, Department of Environmental Health Science, Mailman School of Public Health, Columbia University. February 2015, New York, NY, USA.
3. **Bioclimate and health in Bangladesh – long- and short-term effects of equivalent temperature and precipitation in different subpopulations.** International Research Institute for Climate and Society (IRI), Earth Institute, Columbia University. May 2015, New York, NY, USA.
4. **Do urban green and blue mitigate heat-related excess mortality? – Evidence from Lisbon, Portugal.** 20th International Congress of Biometeorology. September 2015, Cleveland, OH, USA.
5. **The impact of climate and climate change on human health: understanding the role of meteorological drivers, effect modifiers and rising temperatures.** Department of Environmental Health, Rollins School of Public Health, Emory University. September 2016, Atlanta, GA, USA.
6. **Climate change, global change and human health: Direct thermal impacts, effect modifiers and rising temperatures.** Invited Seminar: Climate and Health, Department of Environmental Health Science, Mailman School of Public Health, Columbia University. October 2016, New York, NY, USA.

7. **Health in a changing climate: Analyzing the effects of weather and climate change on human health outcomes.** College of Public Health, University of Nebraska Medical Center. Omaha, NE, USA.
8. **Assessing the evidence of the impact of long-term SNO_2 exposure on incident pediatric asthma: Application of the MR-BRT meta-regression tool.** Annual Conference of the Health Effects Institute. May 2019, Seattle, WA, USA.
9. **The effect of temperature on health: Will climate change bring benefits?** Environmental Health Live Webinar, Department of Environmental and Occupational Health Sciences, School of Public Health, University of Washington. January 28, 2021, Seattle, WA, USA.
10. **Measuring the impact of environmental risk factors on human health: Findings from the Global Burden of Disease study (GBD).** Yale EHS Seminar, Department of Environmental Health Sciences, Yale School of Public Health. April 7, 2021, New Haven, CT, USA.
11. **Climate futures: Estimating the present and future burden of disease attributable to temperature.** HMS Lecture Series, Department of Health Metrics Science, University of Washington School of Medicine. September 17, 2021, Seattle, WA, USA.
12. **Attributing burden of disease to climate change.** Presentation to the IHME Independent Advisory Committee (IAC), National Autonomous University of Mexico (UNAM). November 4–6, 2024, Mexico City, Mexico.
13. **Climate in the Global Burden of Disease study.** Presentation to the GBD Scientific Council. December 11, 2024, International Virtual Meeting.
14. **IHME climate forecasting: Understanding the impact of climate change on human health.** Bill and Melinda Gates Foundation. January 13, 2025, Seattle, WA, USA.
15. **Environmental Risk Assessment within the Global Burden of Disease Study.** Invited Lecture: ENVH Foundations of Environmental and Occupational Health. November 2025, Seattle, WA, USA.
16. **Panelist: “Healthy People Healthy Planet.”** Fundraising event co-hosted by Conservation International (CI) and the Institute for Health Metrics Science (IHME). October 2025, Seattle, WA, USA.

B. INTERNATIONAL

1. **Polycyclic Aromatic Hydrocarbons associated with particulate matter.** Society of German Chemists (GDCh), Environmental & Ecotoxicology Division Annual Meeting. September 2007, Osnabrück, Germany.
2. **Bio-meteorological and air pollution conditions in the megacity of Dhaka, Bangladesh and their effects on public health.** 7th International Conference on Urban Climate. June 2007, Yokohama, Japan.
3. **Health impacts of distorted atmospheric conditions in the megacity of Dhaka.** Satellite Conference: Health in Megacities and Urban Areas, German Epidemiological Association. September 2008, Bielefeld, Germany.
4. **Conceptual framework for assessing the impacts of climate, climate change, and air pollution on public health in Dhaka, Bangladesh.** 27th Annual Meeting of the AK Klima, German Geographical Society. October 2008, Aachen, Germany.
5. **Seasonality of disease in urban poor population groups.** Workshop: Megacities–Megachallenge: Informal Dynamics of Global Change. February 2009, Dhaka, Bangladesh.

6. **The effect of temperature and thermal atmospheric conditions on mortality in Bangladesh.** 7th Conference on Biometeorology, German Meteorological Society. April 2010, Freiburg, Germany.
7. **Investigating the effect of temperature and thermal conditions on all-cause and cardiovascular mortality in Bangladesh using generalized additive models (GAMs).** 29th Annual Meeting of the AK Klima, German Geographical Society. October 2010, Würzburg, Germany.
8. **The effect of atmospheric thermal conditions and urban thermal pollution on mortality in Bangladesh.** BMBF Young Researchers' Symposium: Future Megacities in Balance. October 2010, Essen, Germany.
9. **Bioclimate and Health in urban and rural areas of Bangladesh.** Portuguese-German Workshop on Climate and Health, University of Lisbon. September 2011, Lisbon, Portugal.
10. **The effect of temperature and thermal atmospheric conditions on mortality in Bangladesh.** 19th International Congress on Biometeorology. December 2011, Auckland, New Zealand.
11. **The joint short-term effects of air pollution and meteorological conditions on human all-cause mortality in Berlin and Lisbon.** Urban Environmental Pollution. June 2012, Amsterdam, Netherlands.
12. **Urban-rural differences in heat-related excess mortality in Bangladesh.** International Geographical Congress Pre-Conference Meeting. August 2012, Berlin, Germany.
13. **Using NDVI for estimating the influence and mitigation effect of urban green on heat-related excess mortality in Lisbon, Portugal.** 32nd International Geographical Congress. August 2012, Cologne, Germany.
14. **Modifiers of heat-related excess mortality.** Sino-German Workshop on Health in Chinese Megacities, Chinese Academy of Science. October 2013, Beijing, China.
15. **Climate, Weather and Health: On the bioclimatological trail of Alexander von Humboldt.** Geography Day, Humboldt Universität zu Berlin. May 2014, Berlin, Germany.
16. **Projecting temperature-related mortality in Bangladesh under climate change scenarios.** German Congress for Geography. October 2015, Berlin, Germany.
17. **Leveraging data from the Burden of Disease Study to appraise the potential impact of climate change.** 32nd Annual Conference of the International Society for Environmental Epidemiology. August 2017, Virtual.
18. **Understanding the impact of climate change on human health: IHME's climate forecasting.** New York Climate Week & Lancet Countdown. September 2023, New York, NY, USA.
19. **IHME climate forecasting.** Independent Advisory Committee (IAC) Meeting. November 5–6, 2024, Mexico City, Mexico.
20. **Integrating the impact of climate change into Global Burden of Disease estimates and forecasts.** GBD Scientific Council Meeting. December 11, 2024, International Virtual Meeting.
21. **Invited Presentation: Heat Metrics and Surveillance.** Expert Workshop convened by the World Health Organization (WHO) and the World Meteorological Organization (WMO). June 2025, London, UK.

22. **Keynote: Health effects of extreme temperatures.** Regional Consortium for Climate and Health (CARE-ASIA). October 2025, Singapore.
23. **Invited Presenter and Panelist: Launch of the NUS–IHME Global Burden of Disease Regional Forum.** National University of Singapore. February 2026, Singapore.